

PHYS 350: Honours Electricity and Magnetism

W.R.YCH.

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Abstract

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1 Introduction

Textbook used is David J. Griffiths - Introduction to Electrodynamics Fifth Edition. Course grading is divided as follows:

- 35% problem sets (must be handwritten scans).
- 25% midterm.
- 40% final exam.

2 Vector Analysis

2.1 Terminologies

Definition 1 (Scalar). *A scalar is a quantity with just a magnitude.*

Definition 2 (Vector). *A vector is a quantity with a magnitude and a direction, i.e. $\vec{v} = v_x\hat{x} + v_y\hat{y} + v_z\hat{z}$.*

Definition 3 (Cross Product). *The cross product is a vector perpendicular to both, set by the right hand rule, and it vanishes for parallel vectors:*

$$\vec{v} \times \vec{w} = |\vec{v}||\vec{w}| \sin \theta \hat{n} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{vmatrix} = \hat{x}(v_y w_z - w_y v_z) - \hat{y}(v_x w_z - w_x v_z) + \hat{z}(v_x w_y - w_x v_y).$$

Definition 4 (Dot Product). *The dot product measures how much of $\vec{v}$ points along $\vec{w}$, so it is zero for perpendicular vectors:*

$$\vec{v} \cdot \vec{w} = v_x w_x + v_y w_y + v_z w_z = |\vec{v}||\vec{w}| \cos \theta.$$

Remark 1. Note $\vec{v} \times \vec{w} = -\vec{w} \times \vec{v}$, and $|\vec{v}| = \sqrt{\vec{v} \cdot \vec{v}}$.

Definition 5 (Field). *A field is something with a value at every point in space. A scalar field assigns a number to each point, like temperature $T(x, y, z)$ or the height of a hill. A vector field assigns a vector to each point, like fluid velocity or $\vec{E}$.*

2.2 Differential Calculus

Definition 6 (Del operator). *The del operator*

$$\vec{\nabla} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right).$$

Definition 7 (Gradient). *The gradient turns a scalar into a vector, pointing in the direction of steepest increase of f :*

$$\vec{\nabla} f = \frac{\partial f}{\partial x} \hat{x} + \frac{\partial f}{\partial y} \hat{y} + \frac{\partial f}{\partial z} \hat{z}.$$

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Remark 2. The gradient points in the steepest uphill direction, and its length grows where the hill is steeper. It is always perpendicular to the contours of constant f .

Definition 8 (Divergence). Divergence turns a vector into a scalar, measuring sources and sinks. For a small box, non-zero divergence means the inflow and outflow through its faces do not balance: arrows spreading from a point (positive) mean fluid is created there, arrows converging (negative) mean it is destroyed.

$$\vec{\nabla} \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}.$$

Definition 9 (Curl). Curl turns a vector into a vector, measuring local rotation. Picture a small paddle wheel: if the flow spins it, the curl is non-zero there. Uniform flow does not spin it, but shear flow (parallel arrows, longer on one side) does, so straight field lines do not imply zero curl.

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} = \left(\frac{\partial}{\partial y} F_z - \frac{\partial}{\partial z} F_y, -\left(\frac{\partial}{\partial x} F_z - \frac{\partial}{\partial z} F_x \right), \frac{\partial}{\partial x} F_y - \frac{\partial}{\partial y} F_x \right).$$

Remark 3. The direction of the curl follows the right-hand rule: curl your fingers along the paddle wheel's rotation and your thumb points along $\vec{\nabla} \times \vec{F}$.

Remark 4. Divergence and curl are themselves fields: they can vanish in some places and not others, and a general field has both.

2.3 Integral Calculus

Definition 10 (Integral). An integral is an instruction to move along a region, evaluate something at each step, and add it to a running total.

Definition 11 (Line Integral). A line integral sums $\vec{F}$ along a curve, picking up only the component tangent to the path:

$$\int_C \vec{F} \cdot d\vec{l}.$$

Example. Take $\vec{F} = y\hat{x}$ along a counterclockwise quarter circle of radius R . On the circle $d\vec{l} = R d\phi(-\sin\phi\hat{x} + \cos\phi\hat{y})$ and $y = R\sin\phi$, so

$$\int_C \vec{F} \cdot d\vec{l} = -R^2 \int_0^{\pi/2} \sin^2\phi d\phi = -\frac{R^2\pi}{4}.$$

It is negative because $\vec{F}$ points along $+\hat{x}$ while we travel in the $-\hat{x}$ direction over most of the arc.

Definition 12 (Flux). Flux measures how much of the field pokes through a surface:

$$\int_S \vec{F} \cdot d\vec{a} = \int_S \vec{F} \cdot \hat{n} da.$$

Definition 13 (Volume Integral). *A volume integral sums a scalar over a region, giving the mass if f is a density:*

$$\int_V f d^3\vec{r} = \int_V f dx dy dz.$$

Remark 5. *We write $\oint$ when the curve is a closed loop or the surface is closed.*

2.3.1 The Three Fundamental Theorems

Integrating a derivative over a region reduces to evaluating the original object on the boundary of that region.

Theorem 1 (Gradient theorem). *The boundary of a curve is two points:*

$$\int_a^b \vec{\nabla} h \cdot d\vec{l} = h(b) - h(a).$$

To find my altitude change climbing a hill, I can compare before and after, or add up all the little changes along the way.

Theorem 2 (Stokes' theorem). *The boundary of a surface is a curve:*

$$\int_S (\vec{\nabla} \times \vec{F}) \cdot d\vec{a} = \oint_C \vec{F} \cdot d\vec{l}.$$

To get the total swirl, I can work out the rotation at the boundary, or add up all the little whirlpools inside. Neighbouring whirlpools cancel along their shared edges, leaving only the boundary.

Theorem 3 (Gauss' divergence theorem). *The boundary of a volume is a surface:*

$$\int_V (\vec{\nabla} \cdot \vec{F}) d^3\vec{r} = \oint_S \vec{F} \cdot d\vec{a}.$$

To find the total flux leaking out through the surface, I can add up all the sources inside. Flux between neighbouring cells cancels, leaving only the outer surface.

Definition 14 (Current density). *The current I through a wire is not a local quantity, since a wire has finite thickness. The local version is the current density*

$$\vec{J} = \frac{d\vec{I}}{da_{\perp}},$$

where $da_{\perp}$ is the differential cross section perpendicular to the flow.

Remark 6. *The divergence theorem 3 applied to $\vec{J}$ is how the conservation of charge gets written down.*

2.4 Index Notation

Remark 7. A matrix carries two indices, A_{ij} , where i says which row and j says which column.

Definition 15 (Transpose). The transpose swaps the indices, $(A^T)_{ij} = A_{ji}$.

Definition 16 (Symmetry). A symmetric matrix has $A_{ij} = A_{ji}$, and an anti-symmetric one has $B_{ij} = -B_{ji}$.

Definition 17 (Kronecker delta). The Kronecker delta equals +1 when its condition is fulfilled and 0 otherwise, so it could be the identity matrix in index notation:

$$\delta_{ij} = \begin{cases} +1 & i = j \\ 0 & i \neq j \end{cases}.$$

Example.

$$\sum_j \delta_{ij} v_j = v_i.$$

Proposition 1 (Matrix Multiplication).

$$\vec{w} = A\vec{v} \implies w_i = \sum_j A_{ij} v_j.$$

Proposition 2 (Dot Product).

$$\vec{v} \cdot \vec{w} = \sum_i v_i w_i; \vec{\nabla} \cdot \vec{F} = \sum_i \partial_i F_i.$$

Proposition 3 (Cross Product).

$$(\vec{v} \times \vec{w})_i = \sum_{j,k} \varepsilon_{ijk} v_j w_k; (\vec{\nabla} \times \vec{F})_i = \sum_{j,k} \varepsilon_{ijk} \partial_j F_k.$$

Definition 18 (Levi-Civita symbol). ε_{ijk} is defined by

- $\varepsilon_{123} = +1$, and cyclic permutations keep the sign: $\varepsilon_{123} = \varepsilon_{231} = \varepsilon_{312}$.
- Swapping any two indices flips the sign, so $\varepsilon_{321} = \varepsilon_{213} = \varepsilon_{132} = -1$.
- If any two indices are equal, $\varepsilon_{ijk} = 0$.

Example. For $i = 1$, only the terms with $(j, k) = (2, 3)$ and $(3, 2)$ survive:

$$(\vec{v} \times \vec{w})_1 = \varepsilon_{123} v_2 w_3 + \varepsilon_{132} v_3 w_2 = v_2 w_3 - v_3 w_2.$$

Proposition 4 (Epsilon-delta identity). Contracting two Levi-Civita symbols on a shared index gives

$$\sum_i \varepsilon_{ijk} \varepsilon_{imn} = \delta_{jm} \delta_{kn} - \delta_{jn} \delta_{km}.$$

2.5 Second Derivatives Identities

Definition 19 (Laplacian). *Taking the divergence of a gradient gives*

$$\vec{\nabla} \cdot (\vec{\nabla} f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \equiv \nabla^2 f,$$

where ∇^2 is the Laplacian¹.

Remark 8. ∇^2 also acts on vector fields, component by component: $(\nabla^2 \vec{F})_i = \nabla^2 F_i$.

Example (Sign of the Laplacian). *At a local maximum $\vec{\nabla} f$ points inward from every side, so it has negative divergence and $\nabla^2 f < 0$; at a minimum, $\nabla^2 f > 0$. At a saddle the second derivatives can cancel, so a flat point alone tells you nothing.*

The full list of second derivative identities:

Name	Identity
Divergence of a gradient	$\vec{\nabla} \cdot (\vec{\nabla} f) = \nabla^2 f$
Curl of a gradient	$\vec{\nabla} \times (\vec{\nabla} f) = 0$
Divergence of a curl	$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{F}) = 0$
Curl of a curl	$\vec{\nabla} \times (\vec{\nabla} \times \vec{F}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{F}) - \nabla^2 \vec{F}$

Table 1: Second derivative identities.

2.5.1 Proving the Identities

Curl of a Curl. Set $\vec{G} \equiv \vec{\nabla} \times (\vec{\nabla} \times \vec{F})$. Then,

$$G_i = \sum_{j,k} \varepsilon_{ijk} \partial_j \sum_{m,n} \varepsilon_{kmn} \partial_m F_n = \sum_{j,k,m,n} \varepsilon_{ijk} \varepsilon_{kmn} \partial_j \partial_m F_n.$$

The shared index k is in the third slot of the first symbol, so rotate it forward with the cyclic property $\varepsilon_{ijk} = \varepsilon_{kij}$ to match the proposition (k now first in both):

$$\sum_k \varepsilon_{kij} \varepsilon_{kmn} = \delta_{im} \delta_{jn} - \delta_{in} \delta_{jm}.$$

Reading off the proposition with the summed index k removed: the leftover pairs are (i, j) and (m, n) , giving straight $(\delta_{im} \delta_{jn})$ minus crossed $(\delta_{in} \delta_{jm})$. Hence

$$G_i = \sum_{j,m,n} (\delta_{im} \delta_{jn} - \delta_{in} \delta_{jm}) \partial_j \partial_m F_n.$$

¹Mathematicians often write Δ , but in physics we use ∇^2 .

Each Kronecker delta collapses one sum by setting its indices equal. In the first term δ_{im} forces $m = i$ and δ_{jn} forces $n = j$; in the second, δ_{in} forces $n = i$ and δ_{jm} forces $m = j$:

$$G_i = \sum_j \partial_j \partial_i F_j - \sum_j \partial_j \partial_j F_i.$$

The first term is $\partial_i \sum_j \partial_j F_j = \left(\vec{\nabla}(\vec{\nabla} \cdot \vec{F}) \right)_i$ and the second is $-(\nabla^2 \vec{F})_i$, giving $\vec{\nabla} \times (\vec{\nabla} \times \vec{F}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{F}) - \nabla^2 \vec{F}$. $\square$

Curl of a Gradient. Start from the definition, with $\vec{\nabla} f = \partial_k f$:

$$\left(\vec{\nabla} \times (\vec{\nabla} f) \right)_i = \sum_{j,k} \varepsilon_{ijk} \partial_j \partial_k f.$$

Call this sum S . Since j and k are both summed over, they are dummy labels and can be renamed freely; swap $j \leftrightarrow k$ everywhere:

$$S = \sum_{j,k} \varepsilon_{ikj} \partial_k \partial_j f.$$

Now clean up the two factors. Antisymmetry of ε gives $\varepsilon_{ikj} = -\varepsilon_{ijk}$, and partial derivatives commute so $\partial_k \partial_j f = \partial_j \partial_k f$. Hence

$$S = - \sum_{j,k} \varepsilon_{ijk} \partial_j \partial_k f = -S.$$

So $S = -S$, which forces $S = 0$. Therefore $\vec{\nabla} \times (\vec{\nabla} f) = 0$. $\square$

2.6 The Theory of Vector Fields

Definition 20 (Irrotational and solenoidal). *A field with $\vec{\nabla} \times \vec{F} = 0$ everywhere is irrotational; a field with $\vec{\nabla} \cdot \vec{F} = 0$ everywhere is solenoidal.*

Theorem 4 (Helmholtz). *If $\vec{\nabla} \cdot \vec{F}$ and $\vec{\nabla} \times \vec{F}$ are specified everywhere and $\vec{F} \rightarrow 0$ at infinity, then $\vec{F}$ is uniquely determined. It can be written as $\vec{F} = \vec{\nabla} \psi + \vec{\nabla} \times \vec{A}$, an irrotational part plus a solenoidal part.*

Theorem 5 (Irrotational fields). *Let $\vec{F}$ be a C^1 vector field on a simply connected domain (e.g. all of $\mathbb{R}^3$). The following are equivalent:*

- (i) $\vec{\nabla} \times \vec{F} = 0$ everywhere.
- (ii) $\int_a^b \vec{F} \cdot d\vec{l}$ is independent of path, for any given end points.
- (iii) $\oint_C \vec{F} \cdot d\vec{l} = 0$ for every closed loop C .
- (iv) $\vec{F} = -\vec{\nabla} \psi$ for some scalar field ψ .

Theorem 6 (Solenoidal fields). *Let $\vec{F}$ be a C^1 vector field on a domain with no enclosed voids (e.g. all of $\mathbb{R}^3$). The following are equivalent:*

- (i) $\vec{\nabla} \cdot \vec{F} = 0$ everywhere.
- (ii) $\int_S \vec{F} \cdot d\vec{a}$ is independent of surface, for any given boundary line.
- (iii) $\oint_S \vec{F} \cdot d\vec{a} = 0$ for any closed surface S .
- (iv) $\vec{F} = \vec{\nabla} \times \vec{A}$ for some vector field $\vec{A}$.

Remark 9. *In electrostatics we will find $\vec{\nabla} \times \vec{E} = 0$, which licenses $\vec{E} = -\vec{\nabla}V$ and the electric potential. In magnetostatics we will find $\vec{\nabla} \cdot \vec{B} = 0$, which licenses $\vec{B} = \vec{\nabla} \times \vec{A}$, the vector potential.*

3 Electrostatics

3.1 Electric Field

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Law 1 (Coulomb's law). *The force on a point charge q_2 due to a point charge q_1 is*

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}, \quad (1)$$

where $\hat{r}$ points from q_1 to q_2 .²

Definition 21 (Permittivity of free space). *The permittivity of free space measures how easily an electric field can form in a classical vacuum:*

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N} \cdot \text{m}^2).$$

Definition 22 (Electric field). *The electric field is the force per unit test charge. For a set of point charges q_i ,*

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i^2} \hat{r}_i, \quad \vec{r}_i \equiv \vec{r} - \vec{r}_i. \quad (2)$$

Principle 1 (Superposition). *The field of several sources is the sum of their individual fields, $\vec{E} = \vec{E}_1 + \vec{E}_2 + \dots$*

Remark 10 (Field theory). *Source charges create $\vec{E}$; a test charge Q then feels $\vec{F} = Q\vec{E}$.*

²Coulomb's law has the same shape as Newton's gravity. The reason is geometric rather than a coincidence, and Gauss' law makes that explicit: in d spatial dimensions the flux spreads over a $(d-1)$ -dimensional sphere, so $F \propto 1/r^{d-1}$, and the 4π is the area of the unit sphere in $d=3$.

3.1.1 Continuous Charge Distributions

Definition 23 (Continuous distributions). Replacing q_i by $dq = \lambda dl' = \sigma da' = \rho d\tau'$ turns the sum into an integral:

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r^2} \hat{\mathbf{r}}, \quad \vec{\mathbf{r}} = \vec{r} - \vec{r}', \quad (3)$$

where $\vec{r}$ is the field point and $\vec{r}'$ the source point.

Remark 11. $\hat{\mathbf{r}}$ is not constant: it points differently for every source point $\vec{r}'$, so it cannot come out of the integral. Split it into components first. Direct integration always works (numerically if needed); Gauss' law is quicker but only with symmetry.

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Example (Finite line of charge). A line of length $2L$ along the x axis carries uniform λ . Find $\vec{E}$ at $P = (0, 0, z)$.

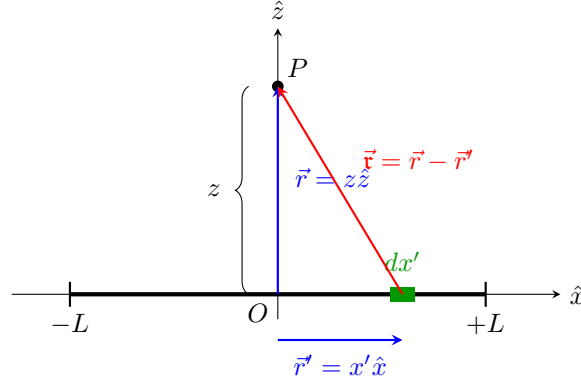


Figure 1: Setup for the finite line of charge. The source point $\vec{r}' = x'\hat{x}$ runs from $-L$ to L ; the field point is P at $\vec{r} = z\hat{z}$; the separation vector is $\vec{\mathbf{r}} = \vec{r} - \vec{r}'$.

With $\vec{r} = z\hat{z}$ and $\vec{r}' = x'\hat{x}$,

$$\vec{\mathbf{r}} = z\hat{z} - x'\hat{x}, \quad r = \sqrt{z^2 + x'^2},$$

so

$$\vec{E} = \frac{\lambda}{4\pi\epsilon_0} \int_{-L}^L \frac{z\hat{z} - x'\hat{x}}{(z^2 + x'^2)^{3/2}} dx'.$$

The $\hat{x}$ term is odd in x' and vanishes by symmetry. Using $\int dx'/(z^2 + x'^2)^{3/2} = x'/(z^2\sqrt{z^2 + x'^2})$,

$$\vec{E} = \frac{\lambda}{2\pi\epsilon_0 z} \cdot \frac{L}{\sqrt{L^2 + z^2}} \hat{z}.$$

- $z \ll L$: $\vec{E} \rightarrow \frac{\lambda}{2\pi\epsilon_0 z} \hat{z}$, the infinite line.
- $z \gg L$: $\vec{E} \rightarrow \frac{Q}{4\pi\epsilon_0 z^2} \hat{z}$ with $Q = 2\lambda L$, a point charge.

Remark 12. Always check limits: a finite distribution should reduce to a simpler known field up close and far away.

3.2 Gauss' Law

Law 2 (Gauss' law). The flux of $\vec{E}$ through any closed surface is set by the charge inside it:

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\epsilon_0}, \quad Q_{enc} = \int_V \rho d\tau. \quad (4)$$

Applying the divergence theorem to the left side gives $\int_V (\vec{\nabla} \cdot \vec{E}) d\tau = \int_V (\rho/\epsilon_0) d\tau$. Since this holds for every volume V , the integrands match, giving the differential form

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}. \quad (5)$$

Remark 13. Flux is a charge detector. Since $d\vec{a}$ always points outward, a field just passing through a closed surface gives negative flux where it enters and positive where it leaves, and these tend to cancel. Net flux needs field lines diverging out of the volume, which needs charge inside. The Gaussian surface is imaginary: nothing is physically there.

Remark 14. To solve for $\vec{E}$, choose a closed surface on which E is constant and $\vec{E}$ is everywhere parallel or perpendicular to $d\vec{a}$, so E comes out of the integral.

3.2.1 Spherical symmetry

Example (Point charge). Surround a point charge q with a sphere S of radius r centred on it. Two facts reduce the integral to a product:

- By symmetry, $\vec{E} = E(r) \hat{r}$: the field is radial and has the same magnitude everywhere on S .
- By geometry, $d\vec{a} = \hat{n} da = \hat{r} da$ on a sphere.

So $\vec{E} \cdot d\vec{a} = E da$, and

$$\oint_S \vec{E} \cdot d\vec{a} = E \underbrace{\oint_S da}_{4\pi r^2} = \frac{q}{\epsilon_0} \implies E = \frac{q}{4\pi\epsilon_0 r^2}.$$

This is Coulomb's law, with $k = 1/4\pi\epsilon_0$.

3.2.2 Cylindrical symmetry

Example (Infinite line of charge). For a line with charge λ per unit length, take a coaxial³ cylinder of radius z and length d . $\vec{E}$ is radial, so the end caps carry no flux:

$$E(2\pi z d) = \frac{\lambda d}{\epsilon_0} \implies E = \frac{\lambda}{2\pi\epsilon_0 z}.$$

The field falls off as $1/z$, not $1/z^2$.

Example (Finite versus infinite line). Near the midpoint, a finite line gives a weaker field than an infinite one with the same λ . The same cylinder encloses the same charge, hence the same total flux, but for the finite line some of it leaks through the end caps, leaving less through the curved surface. The direct integration confirms this: the finite line field is the infinite line field times $L/\sqrt{L^2 + z^2} < 1$.

3.2.3 Planar symmetry

Example (Infinite plane of charge). For a plane with surface charge density σ , take a Gaussian pillbox straddling the plane with caps of area A . By symmetry $\vec{E}$ points straight away from the plane on both sides, so the curved side carries no flux and each cap carries EA :

$$2EA = \frac{\sigma A}{\epsilon_0} \implies E = \frac{\sigma}{2\epsilon_0}.$$

The field does not depend on the distance from the plane.

Example (Infinite plane via Coulomb's law). Use polar coordinates (r', ϕ') on the plane, centred below the field point at height z . Each piece $\sigma r' dr' d\phi'$ pushes along $\hat{r}$. Sideways components cancel in pairs, so keep only the $\hat{z}$ part via $\cos\psi = z/\sqrt{r'^2 + z^2}$:

$$\vec{E} = \hat{z} \int_0^{2\pi} d\phi' \int_0^\infty \frac{\sigma}{4\pi\epsilon_0} \frac{r' z}{(r'^2 + z^2)^{3/2}} dr' = \frac{\sigma z}{2\epsilon_0} \left[-\frac{1}{\sqrt{r'^2 + z^2}} \right]_0^\infty \hat{z} = \frac{\sigma}{2\epsilon_0} \hat{z}.$$

Why distance drops out: up close each piece pushes hard but mostly at a bad angle; far away each piece is weak but more of them push at a good angle. The two effects cancel exactly. For a line, stepping back also brings in more charge at good angles, but only $\propto r$, not enough to beat $1/r^2$, hence $E \propto 1/r$.

3.2.4 Why the fall-off depends on the geometry

The exponent in the fall-off counts how much charge a cone of field lines can “see” as it grows. Picture a narrow cone opening from the observation point toward the source: its cross section at distance r grows like r^2 , which alone gives the $1/r^2$ dilution of a point charge. For extended sources, the charge inside the cone grows too:

³Sharing a common central axis.

- Line of charge: the cone intercepts a length $\propto r$, so $E \propto r/r^2 = 1/r$.
- Sheet of charge: the cone intercepts an area $\propto r^2$, so $E \propto r^2/r^2$, a constant. The infinite sheet has a uniform field.

Remark 15. *For two equal positive charges, no field lines cross the midplane: the field there is parallel to the plane and vanishes at the midpoint. Gauss' law still holds, but without symmetry it cannot solve for $\vec{E}$.*

3.3 The Curl of $\vec{E}$

Example (Line integral of a point charge). *For a point charge at the origin, $d\vec{l} = dr \hat{r} + r d\theta \hat{\theta} + r \sin \theta d\phi \hat{\phi}$ and only the radial part survives:*

$$\int_a^b \vec{E} \cdot d\vec{l} = \frac{q}{4\pi\epsilon_0} \int_{r_a}^{r_b} \frac{dr}{r^2} = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r_a} - \frac{1}{r_b} \right).$$

This depends only on the endpoints, so around a closed loop ($r_a = r_b$) it vanishes.

Theorem 7 (Electrostatic fields are curl-free). *For any electrostatic field, $\oint \vec{E} \cdot d\vec{l} = 0$ and $\vec{\nabla} \times \vec{E} = 0$.*

Proof. By superposition, any static $\vec{E}$ is a sum of point-charge fields, each with zero loop integral. Stokes' theorem then gives $\int_S (\vec{\nabla} \times \vec{E}) \cdot d\vec{a} = 0$ for every surface, so $\vec{\nabla} \times \vec{E} = 0$. $\square$

Remark 16. *Physically, $Q \oint \vec{E} \cdot d\vec{l}$ is the work done on a test charge around a closed loop. If it were nonzero, a charge could gain energy just by circling, violating energy conservation.*

3.4 Electric Potential

Definition 24 (Electric potential). *Since $\vec{\nabla} \times \vec{E} = 0$, the irrotational fields theorem guarantees a scalar V with*

$$\vec{E} = -\vec{\nabla}V.$$

The minus sign is convention: positive charges are pushed toward lower potential energy.

Remark 17. $\oint \vec{E} \cdot d\vec{l} = 0$ holds only in electrostatics. If fields vary in time, a charge can gain net energy around a loop.

Proposition 5 (Potential from the field).

$$V(\vec{r}) = - \int_{ref}^{\vec{r}} \vec{E} \cdot d\vec{l}.$$

The path does not matter since the integral is path-independent⁴. Changing the reference point adds a constant to V , which leaves $\vec{E} = -\vec{\nabla}V$ unchanged (gauge freedom). By convention the reference is at infinity, unless the charge distribution extends to infinity.

Example (Point charge). Along a radial path,

$$V(r) = -\frac{q}{4\pi\epsilon_0} \int_{\infty}^r \frac{dr'}{r'^2} = \frac{q}{4\pi\epsilon_0 r}.$$

Proposition 6 (Potential of a distribution). By superposition,

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{r}')}{|\vec{r} - \vec{r}'|} d\tau',$$

where $\rho(\vec{r}')d\tau' = dq$. Unlike the integral for $\vec{E}$, there is no $\hat{r}$, so computing V first and then $\vec{E} = -\vec{\nabla}V$ is often easier.

Law 3 (Poisson's equation). Combining $\vec{\nabla} \cdot \vec{E} = \rho/\epsilon_0$ with $\vec{E} = -\vec{\nabla}V$,

$$\nabla^2 V = -\frac{\rho}{\epsilon_0}.$$

3.5 Work and Energy in Electrostatics

Proposition 7 (Potential energy of a test charge). Bringing Q in from infinity against the field takes

$$W = -Q \int_{\infty}^{\vec{r}} \vec{E} \cdot d\vec{l} = QV(\vec{r}).$$

So V is potential energy per unit charge, in $\text{J/C} = \text{V}$.

Proposition 8 (Energy of N point charges). The first charge costs no work, since no field exists yet, and each later one feels the potential of those already placed, giving one term per pair:

$$U = \frac{1}{2} \sum_{i \neq j} \frac{q_i q_j}{4\pi\epsilon_0 |\vec{r}_i - \vec{r}_j|},$$

where the $\frac{1}{2}$ corrects for counting each pair twice.

Remark 18. U does not obey superposition: fields add, but energy goes like E^2 , so combining two configurations gives $U = U_1 + U_2 + (\text{interaction cross terms})$.

Theorem 8 (Energy stored in the field). For a continuous distribution,

$$U = \frac{1}{2} \int \rho V d\tau = \frac{\epsilon_0}{2} \int_{\text{all space}} E^2 d\tau, \quad u = \frac{\epsilon_0}{2} E^2 \text{ (energy density)}.$$

⁴Since $\vec{\nabla} \times \vec{E} = 0$, statement (ii) of the irrotational fields theorem applies.

Proof. The pair sum can be written $U = \frac{1}{2} \sum_i q_i V(\vec{r}_i)$, where $V(\vec{r}_i)$ is the potential at q_i due to the others. For a continuous distribution this becomes $U = \frac{1}{2} \int \rho V d\tau$. Substituting $\rho = -\epsilon_0 \nabla^2 V$ from Poisson's equation,

$$U = -\frac{\epsilon_0}{2} \int (\nabla^2 V) V d\tau.$$

By the product rule in index notation,

$$\sum_i \partial_i (V \partial_i V) = \sum_i (\partial_i V)^2 + V \sum_i \partial_i \partial_i V \implies (\nabla^2 V) V = \vec{\nabla} \cdot (V \vec{\nabla} V) - |\vec{\nabla} V|^2.$$

So

$$U = \frac{\epsilon_0}{2} \int |\vec{\nabla} V|^2 d\tau - \frac{\epsilon_0}{2} \int \vec{\nabla} \cdot (V \vec{\nabla} V) d\tau.$$

The divergence theorem turns the second integral over all space into a flux through its boundary, a sphere at infinity: $\oint V \vec{\nabla} V \cdot d\vec{a}$. For a bounded distribution $V \sim 1/r$ and $|\vec{\nabla} V| \sim 1/r^2$ while the area grows as r^2 , so this term falls off as $1/r$ and vanishes. With $\vec{E} = -\vec{\nabla} V$, $U = \frac{\epsilon_0}{2} \int E^2 d\tau$. $\square$

Remark 19 (Self-energy). *The discrete sum excludes $i = j$; the continuous integral does not. So $\frac{\epsilon_0}{2} \int E^2 d\tau$ is always positive, while the pair sum for two opposite charges is negative. The difference is the infinite self-energy of point charges.*

4 Potentials

4.1 Multipole Expansion

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Example (Two charges on the z axis). Place q_1 at $z = d/2$ and q_2 at $z = -d/2$. With θ measured from the z axis,

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{\mathbf{r}_1} + \frac{q_2}{\mathbf{r}_2} \right), \quad \mathbf{r}_{1,2}^2 = r^2 + \left(\frac{d}{2} \right)^2 \mp rd \cos \theta.$$

Writing $1/\mathbf{r}_{1,2} = \frac{1}{r} (1 + \epsilon_{\mp})^{-1/2}$ with $\epsilon_{\mp} = \frac{d}{2r} \left(\frac{d}{2r} \mp 2 \cos \theta \right)$ and expanding for $d \ll r$, $(1 + \epsilon)^{-1/2} = 1 - \frac{1}{2}\epsilon + \frac{3}{8}\epsilon^2 - \dots$, gives

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left[\underbrace{\frac{q_1 + q_2}{r}}_{\text{monopole}} + \underbrace{\frac{(q_1 - q_2)d \cos \theta}{2r^2}}_{\text{dipole}} + \underbrace{\frac{(q_1 + q_2)d^2}{4} \frac{3 \cos^2 \theta - 1}{2r^3}}_{\text{quadrupole}} + \dots \right].$$

Each term gains a power of $1/r$, so $E \sim 1/r^2, 1/r^3, 1/r^4, \dots$

Remark 20. Setting $q_1 = -q_2$ kills the monopole term, leaving $V \sim 1/r^2$ and $E \sim 1/r^3$ at leading order.

Definition 25 (Ideal dipole). *The n -th term carries $q d^n$. Taking $q \rightarrow \infty$, $d \rightarrow 0$ with qd fixed kills every term beyond the dipole, leaving a point object with dipole moment $\vec{p} = q\vec{d}$ and exactly $V \sim 1/r^2$.*

Remark 21. *Two opposite dipoles placed close together cancel at dipole order, giving a quadrupole with $V \sim 1/r^3$. In general the pattern continues: two opposite 2^n -poles make a 2^{n+1} -pole.*

Theorem 9 (Multipole expansion). *For r large compared with the size of the distribution,*

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{Q}{r} + \frac{\vec{p} \cdot \hat{r}}{r^2} + \frac{\sum_{i,j} \hat{r}_i Q_{ij} \hat{r}_j}{r^3} + \dots \right),$$

with

$$Q = \int \rho(\vec{r}') d\tau', \quad \vec{p} = \int \vec{r}' \rho(\vec{r}') d\tau', \quad Q_{ij} = \int \frac{1}{2} (3r'_i r'_j - r'^2 \delta_{ij}) \rho(\vec{r}') d\tau'.$$

Equivalently (Griffiths' form), with α the angle between $\vec{r}$ and $\vec{r}'$ and P_n the Legendre polynomials,

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \sum_{n=0}^{\infty} \frac{1}{r^{n+1}} \int (r')^n P_n(\cos \alpha) \rho(\vec{r}') d\tau'.$$

$P_0 = 1$ gives the monopole, $P_1 = \cos \alpha$ the dipole (since $r' \cos \alpha = \vec{r}' \cdot \hat{r}$), and $P_2 = \frac{1}{2}(3 \cos^2 \alpha - 1)$ the quadrupole.

Remark 22. *This is a series in $1/r$, good at large r , the opposite of a Taylor series in r .*

Remark 23 (Origin dependence). *Shifting the origin by $\vec{a}$ changes $\vec{p} \rightarrow \vec{p} - Q\vec{a}$, so $\vec{p}$ is well defined only when $Q = 0$ (a neutral molecule, not an ion). In general only the first non-vanishing moment is origin-independent.*

5 Electric Fields in Matter

6 Magnetostatics

7 Magnetic Fields in Matter

8 Electrodynamics

A Appendix

Remark 24 (Compactified dimensions). *Suppose an extra dimension exists but is curled up, the way a flat sheet with coordinates (x, y) can be rolled into a*

cylinder whose second coordinate w is periodic with a small radius. At distances much smaller than that radius, field lines spread out into all the dimensions and the field falls off faster than $1/r^2$. At distances much larger than it, there is only room to spread out in the large dimensions and the usual $1/r^2$ is recovered. So a measured deviation from the inverse square law at short range would be evidence for extra dimensions.

This has been tested for gravity by the Eöt-Wash torsion balance experiments (2004, 2007, 2020), which checked the $1/r^2$ law down to separations of $52\,\mu\text{m}$. Newtonian gravity fit the data, ruling out gravitational strength Yukawa interactions with ranges above $38.6\,\mu\text{m}$ at 95% confidence.

B Useful Links